Derivatives (Univariate): A Rigorous Introduction

Math Foundations Map — Node 12

1 Motivation

Given a function $f : I \to \mathbb{R}$ on an open interval I and a point $c \in I$, we wish to assign a number $f'(c)$ that captures the *instantaneous rate of change* of f at c. Geometrically, this number is the slope of the line that best approximates the graph of f in an arbitrarily small neighbourhood of $(c, f(c))$. Algebraically, it is the limit of the slopes of secant lines through $(c, f(c))$ as the second point collapses onto $(c, f(c))$.

2 Definitions

Definition 1 (Derivative at a point)**.** Let $f : I \to \mathbb{R}$ where $I \subseteq \mathbb{R}$ is an open interval, and let $c \in I$. We say f is **differentiable at** c if the limit

$$f'(c) \;:=\; \lim_{h \to 0} \frac{f(c+h) - f(c)}{h}$$

exists in $\mathbb{R}$. The value $f'(c)$ is called the **derivative** of f at c. An equivalent formulation, obtained by substituting $x = c + h$, is

$$f'(c) \;=\; \lim_{x \to c} \frac{f(x) - f(c)}{x - c}.$$

Definition 2 (Differentiability on an interval)**.** f is **differentiable on** I if $f'(c)$ exists for every $c \in I$. In this case the assignment $c \mapsto f'(c)$ defines a function $f' : I \to \mathbb{R}$, called the **derivative function**.

Definition 3 (Local linear approximation)**.** f is differentiable at c with derivative m if and only if there exists a function ε defined on a neighbourhood of 0 with $\lim_{h \to 0} \varepsilon(h) = 0$ and

$$f(c+h) \;=\; f(c) + m \cdot h + h \cdot \varepsilon(h).$$

3 Differentiability implies continuity

The first nontrivial fact about differentiation is that it strengthens continuity. We isolate this as the main theorem of the lesson.

Theorem 4 (Differentiable implies continuous)**.** *If $f : I \to \mathbb{R}$ is differentiable at $c \in I$, then f is continuous at c.*

Proof. We must show $\lim_{x \to c} f(x) = f(c)$, equivalently $\lim_{x \to c}(f(x) - f(c)) = 0$. For $x \in I$ with $x \neq c$, we use the algebraic identity

$$f(x) - f(c) \;=\; \frac{f(x) - f(c)}{x - c} \cdot (x - c). \tag{1}$$

By the definition of $f'(c)$,

$$\lim_{x\to c} \frac{f(x)-f(c)}{x-c} \;=\; f'(c) \;\in\; \mathbb{R}.$$

Also, trivially,

$$\lim_{x\to c}(x-c) \;=\; 0.$$

The product rule for limits (which holds whenever both factors have finite limits) applied to (1) yields

$$\lim_{x\to c}\bigl(f(x)-f(c)\bigr) \;=\; \left(\lim_{x\to c}\frac{f(x)-f(c)}{x-c}\right)\cdot\left(\lim_{x\to c}(x-c)\right) \;=\; f'(c)\cdot 0 \;=\; 0.$$

Therefore $\lim_{x\to c} f(x) = f(c)$, i.e. f is continuous at c. □

Remark 5. The converse is false. The function $f(x) = |x|$ is continuous at 0 but the two one-sided limits of the difference quotient are $+1$ and -1, so $f'(0)$ does not exist.

4 Differentiation rules

Proposition 6 (Algebraic rules)**.** *Let f, g be differentiable at c, and let $\alpha \in \mathbb{R}$. Then:*

1. $(\alpha f)'(c) = \alpha f'(c)$;
2. $(f+g)'(c) = f'(c) + g'(c)$;
3. $(fg)'(c) = f'(c)g(c) + f(c)g'(c)$;
4. *if* $g(c) \neq 0$, $(f/g)'(c) = \dfrac{f'(c)g(c) - f(c)g'(c)}{g(c)^2}$.

Proposition 7 (Chain rule)**.** *If g is differentiable at c and f is differentiable at $g(c)$, then $f \circ g$ is differentiable at c and $(f \circ g)'(c) = f'(g(c)) \cdot g'(c)$.*

5 The Mean Value Theorem

Theorem 8 (Rolle)**.** *If $f : [a,b] \to \mathbb{R}$ is continuous on $[a,b]$, differentiable on (a,b), and $f(a) = f(b)$, then there exists $\xi \in (a,b)$ with $f'(\xi) = 0$.*

Theorem 9 (Mean Value Theorem)**.** *If $f : [a,b] \to \mathbb{R}$ is continuous on $[a,b]$ and differentiable on (a,b), then there exists $\xi \in (a,b)$ with*

$$f'(\xi) \;=\; \frac{f(b)-f(a)}{b-a}.$$

The MVT is the bridge between local information (f') and global behaviour ($f(b) - f(a)$). Consequences include: monotonicity criteria, L^∞ bounds on increments, Taylor's theorem with Lagrange remainder, and the uniqueness of antiderivatives up to additive constants.

6 Example: $\frac{d}{dx}x^2$ from first principles

Example 10. For $f(x) = x^2$ and $c \in \mathbb{R}$,

$$\frac{f(c+h)-f(c)}{h} = \frac{(c+h)^2 - c^2}{h} = \frac{2ch + h^2}{h} = 2c + h \;\xrightarrow{h\to 0}\; 2c.$$

Hence $f'(c) = 2c$, i.e. $\frac{d}{dx}x^2 = 2x$.